FIN41360: PORTFOLIO & RISK MANAGEMENT

Group Work Assignment 2

Risk Modelling & Analysis

Group #35

Date: 30/04/2026

We declare that all material included in this project is the result of our own work and that due acknowledgement has been given in the bibliography and references to all sources be they printed, electronic or personal.

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1 Introduction

Financial institutions are required to quantify and report portfolio risk exposures to regulatory bodies and internal governance committees with precision and transparency. This report fulfils that function, presenting a comprehensive risk measurement analysis for a Global Investment Bank's Board Risk Committee across three separate $1,000,000 investments: a pure equity portfolio, an interest rate portfolio, and a combined 60/40 allocation.

Risk estimates are produced using four distinct methodologies: Parametric, Cornish-Fisher, Historical Simulation, and Monte Carlo while being supplemented by an Exponential Spectral Risk Measure, across both 99% and 95% confidence levels. The reliability of each model is evaluated through a rolling backtesting framework, with sensitivity analysis supporting the final recommendations. This report examines the performance of each methodology across all three portfolios, drawing on both academic literature and empirical backtesting to arrive at a justified risk measure recommendation for each investment.

2 Literature Review

While VaR remains the regulatory benchmark under Basel III, its theoretical limitations are well-documented. Artzner *et al.* (1999) demonstrated VaR fails the subadditivity axiom and proposed the framework of coherent risk measures, of which Expected Shortfall is the canonical example. ES captures the average loss beyond the VaR threshold rather than merely identifying it. This theoretical superiority of ES over VaR motivates its inclusion as a primary measure throughout this report.

A central challenge in risk modelling is the distributional assumption underpinning parametric methods. Mandelbrot (1963) and Fama (1965) established that financial returns exhibit fat tails and excess kurtosis, departing from normality. Engle's (1982) ARCH framework demonstrated that large return shocks tend to cluster when modelling time-varying volatility. This literature motivates the use of distribution-adjusted and non-parametric methods alongside the parametric baseline.

The Cornish-Fisher (1938) expansion provides a distributional correction by incorporating empirical skewness and excess kurtosis, though Maillard (2012) showed that the polynomial approximation underpinning the CF expansion becomes unreliable under extreme kurtosis, a relevant limitation given the equity portfolio's kurtosis of 7.149 documented in this report. Historical Simulation avoids distributional assumptions entirely by treating each observed return as equally probable, capturing fat tails and skewness from the empirical distribution, though its equal weighting of all observations is a recognised lim-

itation during periods of changing volatility regimes. Spectral risk measures, introduced by Acerbi and Tasche (2002), address this by weighting losses according to a risk aversion function across the full distribution, with Cotter and Dowd (2007) providing the foundation for the exponential SRM implemented here.

For backtesting, Kupiec (1995) and Christoffersen (1998) provide the unconditional coverage and independence tests used to evaluate model performance. Portfolio construction follows Markowitz (1952), with equal weighting motivated by DeMiguel *et al.* (2009), who showed naïve 1/N allocation consistently matches or outperforms mean-variance optimisation under estimation error.

3 Portfolio Construction

The portfolio was allocated as 60% Equity and 40% bonds, using the 30 constituents of the Dow Jones Industrial Average (DJIA) (Appendix 1) and the Bloomberg Aggregate Bond Index (LBUSTRUU) as a proxy for the interest rate portfolio. This weighting was justified based on both an empirical and practicality review. The 60/40 split reflects the traditional domestic US benchmark widely used by institutional portfolio managers adopted by Chopra *et al* (1993) as their passive benchmark on the basis of established long-term risk-return characteristics. Equal weighting across the 30 DJIA constituents was adopted in preference to mean-variance optimisation, consistent with the findings of DeMiguel *et al* (2009). They demonstrated across seven empirical datasets that no optimising model consistently outperforms the naïve 1/N rule owing to estimation error in means and covariance, a problem coined by Michaud (1989) as error maximisation. This was chosen over Global Minimum Portfolio optimisation as DeMiguel *et al* (2009) also found this method adds complexity without adding insight. Additionally, the sample size constraint of this report would result in weights which are extreme and unstable which was also noted by Michaud (1989).

4 Data

All data was sourced using Bloomberg terminals covering 28th March 2022 to 27th March 2026, covering 1,000 common trading days after aligning different equity and bond trading calendars. The equity data comprises the 30 current DJIA constituents and interest rate data using the LBUSTRUU as a proxy for broad fixed income exposure. Log returns were then computed as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) \tag{1}$$

Log returns are preferred over simple returns because they are time-additive.

4.1 Testing Assumptions

The normality assumption is assessed through Figures 1 and 2.

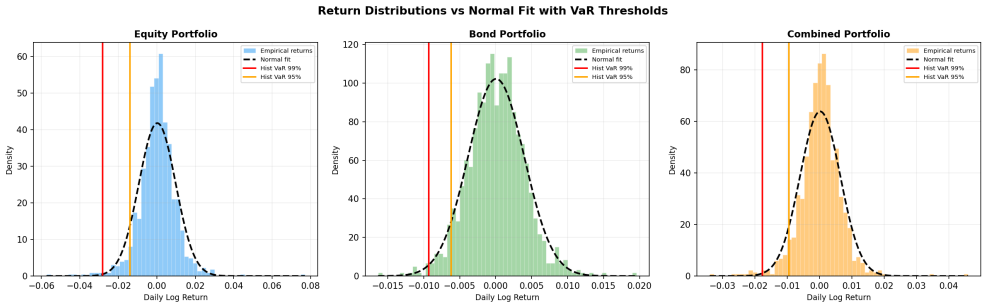


Figure 1: Returns Distributions

The equity portfolio exhibits a sharper central peak and fatter tails than the normal fitted curve, while the bond portfolio tracks closer to normality, suggesting the assumption is more defensible for interest rate risk.

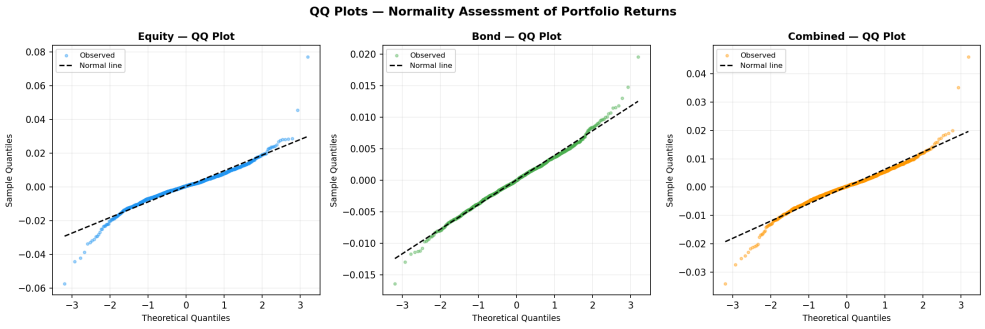


Figure 2: QQ Plots of Normality

The QQ plots corroborate this. The equity portfolio exhibits S-shaped deviation at both tails, with the left extreme reaching -0.06, visualising how extreme losses occur more frequently than the normal distribution implies. The bond portfolio tracks the reference line closely, and the combined portfolio shows meaningful left-tail deviation driven by the equity allocation.



Figure 3: Daily Log Returns of the Three Portfolios

Figure 3 reveals volatility clustering across all three portfolios during the shaded 2022 stress period, consistent with the ARCH effects documented by Engle (1982). These effects show that returns are not i.i.d. and are serially correlated. A large volatility spike is visible around April 2025, in line with the "Liberation Day" tariff shock. This type of event represents the type of fat-tail occurrence that the normal distribution underestimates, reinforcing the case for non-parametric approaches that capture such events from the empirical distribution.

Collectively, the data only partially satisfies the parametric assumptions. Parametric estimates should be interpreted as a lower bound on tail risk for the equity and combined portfolios, with non-parametric and distribution-adjusted methods expected to produce higher estimates.

5 Methodology and Expectations

5.1 Parametric Approach: VaR and ES

The Parametric Approach assumes portfolio returns follow a normal distribution with mean μ and standard deviation σ estimated from the historical return series. All estimates are computed at 99% and 95% confidence levels, with 99% serving as the primary

estimate, consistent with Basel III internal model requirements.

Parametric VaR is computed as:

$$VaR_\alpha = (-\mu + z_\alpha \sigma)\, W \tag{2}$$

where z_α is the standard normal quantile and W is the \$1,000,000 investment. ES extends this analytically as:

$$ES_\alpha = \left(-\mu + \sigma \frac{\phi(z_\alpha)}{\alpha}\right) W \tag{3}$$

where $\phi(z_\alpha)$ is the standard normal probability density function evaluated at z_α and α is the significance level. Unlike VaR, ES is a coherent risk measure satisfying subadditivity (Artzner *et al* 1999), making it theoretically superior for risk aggregation. The distributional violations previously identified motivate the distribution-adjusted and non-parametric methods that follow, with parametric methods providing a benchmark.

5.2 Cornish-Fisher VaR and ES

The Cornish-Fisher (CF) approach extends the parametric method by modifying the z-score to account for empirical skewness and excess kurtosis, producing a distribution-adjusted quantile that better captures fat tails. The adjusted quantile is derived using the Cornish-Fisher expansion:

$$z_{CF} = z + \frac{(z^2-1)s}{6} + \frac{(z^3-3z)k}{24} - \frac{(2z^3-5z)s^2}{36} \tag{4}$$

where s is the empirical skewness and k is the excess kurtosis. Positive excess kurtosis increases z_{CF}, pushing the VaR further into the tail and producing higher loss estimates than the standard parametric approach.

CF ES is computed by integrating the CF-adjusted quantile function across the tail region using 500 equally spaced probability points. This produces the average CF-adjusted loss across all tail scenarios, similar to the ES formula under normality but accounting for distributional adjustments introduced by skewness and kurtosis. The CF expansion is known to be unreliable under extreme kurtosis (Maillard, 2012), producing implausible estimates at lower confidence levels. The CF approach is reliant on sample estimates of skewness and kurtosis, which are subject to estimation error, necessitating a non-parametric approach.

5.3 Historical Simulation VaR and ES

Historical Simulation (HS) is a non-parametric approach treating each of the 1,000 observed daily log returns as an equally likely outcome for the following day, eliminating distributional misspecification risk. Historical VaR corresponds to the empirical quantile at the chosen confidence level. At 99% with 1,000 observations, this is the 10th daily return. Historical ES averages all returns below this threshold meaning 10 observations drive the 99% ES estimate making it sensitive to a small number of extreme events.

The equal weighting of all observation means results in extreme values carrying the same weight as calmer periods such as the Liberation Day spike, which is a recognised limitation of the approach.

5.4 Monte Carlo VaR and ES

MC simulation generates 100,000 scenarios of future returns by drawing from a bivariate normal distribution estimated from the empirical covariance matrix of equity and bond returns, preserving their observed correlation. VaR is the relevant simulated quantile and ES is the mean of losses beyond it. As the simulation draws from a normal distribution, MC results will closely track parametric estimates. This acknowledged limitation means MC serves primarily as a verification of parametric results rather than an independent risk measure.

5.5 Exponential Spectral Risk Measure

Spectral risk measures generalise ES by weighting losses across the full return distribution according to a risk aversion function rather than applying equal weight beyond a fixed confidence threshold, making them sensitive to the entire shape of the loss distribution (Acerbi and Tasche, 2002).

The exponential specification applies the weighting function:

$$\phi(p) = \frac{\exp\left(\frac{1-p}{\gamma}\right)}{\gamma\left(\exp\left(\frac{1}{\gamma}\right) - 1\right)} \tag{5}$$

where p is the probability level and γ is the risk aversion parameter. γ is set to 0.2, reflecting moderate risk aversion consistent with the framework illustrated by Cotter and Dowd, (2007). However, this report conducts a sensitivity analysis to examine the effects of adjusting γ. The integral is approximated using 1,000 probability slices, confirmed by convergence analysis. The spectral measure integrates over the full distribution rather

than conditioning on tail exceedances, producing lower absolute figures than ES. This is a feature of the measure rather than a deficiency.

5.6 Volatility and Maximum Drawdown

Annualised volatility measures return dispersion scaled by $\sqrt{252}$, providing a standardised measure of overall portfolio risk:

$$\sigma_{\text{ann}} = \sigma_{\text{daily}}\sqrt{252} \tag{6}$$

Maximum drawdown captures the largest peak-to-trough decline over the sample period:

$$MDD = \min_{t}\left(\frac{V_t - \max_{\tau \le t} V_\tau}{\max_{\tau \le t} V_\tau}\right) \tag{7}$$

where V_t is the portfolio value at time t. This quantifies the worst realised capital loss an investor would have experienced, complementing the forward-looking VaR and ES estimates with historical loss experience.

6 Equity Portfolio Results

Table 1: Risk Estimates Summary for the One-Day Horizon as at 27 March 2026 using a $1,000,000 Investment

Measure	**Equity ($)**	**Bond ($)**	**Combined ($)**
Parametric VaR (99%)	21,964.00	9,017.84	14,345.64
Parametric ES (99%)	25,202.13	10,340.95	16,462.35
Cornish-Fisher VaR (99%)	37,977.62	10,328.04	22,706.72
Cornish-Fisher ES (99%)	58,651.89	12,100.32	33,702.07
Historical VaR (99%)	28,165.79	9,286.91	17,691.04
Historical ES (99%)	37,160.69	11,414.58	23,825.85
Monte Carlo VaR (99%)	22,034.22	8,792.19	14,376.22
Monte Carlo ES (99%)	25,154.18	10,342.79	16,369.04
Exponential SRM ($\gamma = 0.2$)	9,651.47	4,076.99	6,360.59
Parametric VaR (95%)	15,451.80	6,356.93	10,088.71
Parametric ES (95%)	19,444.76	7,988.47	12,698.85
Cornish-Fisher VaR (95%)	14,090.19	6,375.90	9,408.13
Cornish-Fisher ES (95%)	29,525.18	8,421.83	17,872.44
Historical VaR (95%)	13,845.55	6,197.81	9,497.42
Historical ES (95%)	22,671.80	8,176.41	14,534.47
Monte Carlo VaR (95%)	15,597.46	6,368.60	10,173.36
Monte Carlo ES (95%)	19,549.30	7,980.00	12,746.31
Daily Volatility	0.9956%	0.3905%	0.6246%
Annualised Volatility	15.1693%	6.1982%	9.9159%
Maximum Drawdown	-22.2126%	-11.5754%	-17.1299%

The equity portfolio produces the highest risk estimates across every measure and both confidence levels, consistent with the established risk-return trade-off in equity markets (Asness, 2000). At 99% confidence, historical VaR of $28,166 and historical ES of $37,161 are the primary non-parametric estimates, both substantially above their bond and combined counterparts, illustrating the higher risk premium associated with equity markets.

The key driver of elevated equity risk is the pronounced departure from normality. Skewness of 0.006 and excess kurtosis of 7.149 confirmed significant fat tails. The practical consequence is visible in the CF vs parametric comparison: CF VaR of $37,978 exceeds parametric VaR of $21,964 by 72.9% and CF ES of $58,652 is more than double parametric ES of $25,202. This divergence quantifies the cost of the normality assumption. A risk manager relying solely on parametric estimates would allocate $6,202 less capital per $1,000,000 invested than the empirical distribution requires thus leaving the portfolio undercapitalised. While CF ES is credible at 99% (where kurtosis adjustment functions as intended) it becomes unreliable at 95% where the polynomial approximation breaks down. Thus, these figures will be excluded from cross-method comparison.

Historical VaR exceeds parametric by $6,202 (28.24%) at 99%, further evidence of the cost of imposing normality on a fat-tailed distribution. MC VaR and ES track parametric within 1% confirming that simulation adds no independent tail risk information when both assume normality.

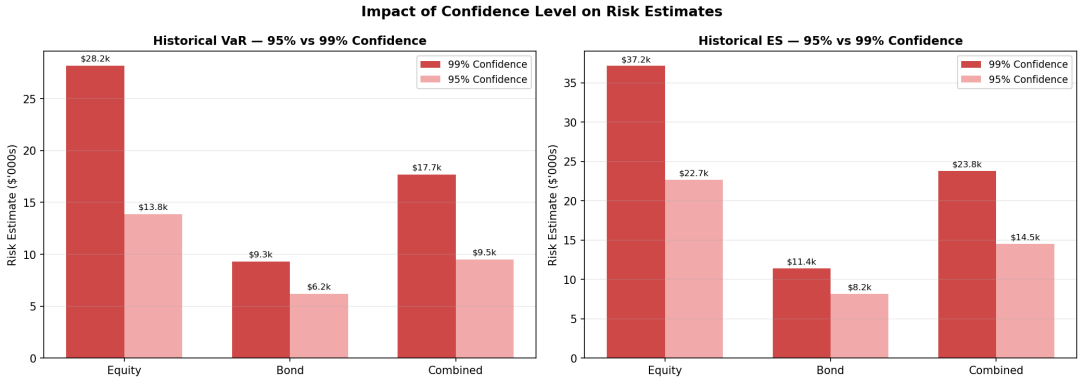


Figure 4: Historical VaR and ES for each portfolio at 95% and 99% confidence levels

Equity is uniquely sensitive to confidence level. Historical VaR more than doubles from $13,846 at 95% to $28,166 at 99%, confirming that fat tails are a tail-specific phenomenon.

Complementary measures reinforce this. Annualised volatility of 15.17% and maximum drawdown of -22.21% are both the highest across the three portfolios, with the drawdown figure nearly double the bond equivalent. This reflects the severity of the 2022 bear market and the April 2025 "Liberation Day" shock on an undiversified equity position.

7 Bond Portfolio Results

The bond portfolio generates the lowest risk estimates across every measure and both confidence levels, consistent with the defensive characteristics of fixed-income securities (Asness, 2000). At 99% confidence, historical VaR of $9,287 and ES of $11,415 represent decreases of $18,879 and $25,746 respectively from equity counterparts, quantifying the stabilising role of bonds in a multi-asset portfolio.

The defining characteristic of the bond portfolio is near-normality. Kurtosis of 1.141 is substantially below equity's 7.149 and close to the theoretical normal value of 3. Thus, the normality assumption is more defensible here than for equity or combined. This is evidenced by method convergence: CF VaR of $10,328 exceeds parametric VaR of $9,018 by only 14.53%, compared to 72.9% divergence for equity. The gap between parametric and historical VaR collapses to just $269 at 99%. A risk manager relying solely on parametric estimates would allocate only $269 less capital per $1,000,000 than the empirical distribution requires.

Complementary measures reinforce this. Annualised volatility of 6.20% and maximum drawdown of -11.58% are both the lowest across the three portfolios. The drawdown figure is nearly half the equity equivalent, reflecting the comparative stability of bond returns even through the 2022 rate shock.

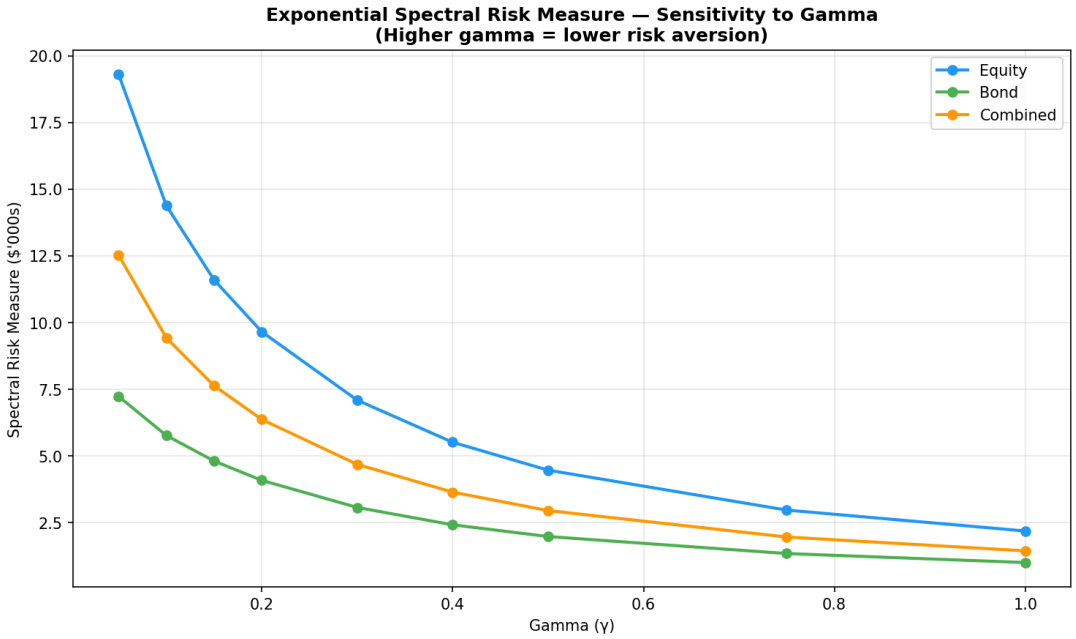


Figure 5: Exponential Spectral Risk Measure against Varying Levels of Gamma

Figure 5 illustrates this stability through spectral risk measure sensitivity analysis. Bond produces the lowest SRM at every gamma value and exhibits the lowest slope across the range. This confirms that bond risk estimates are least sensitive to the risk aversion parameter and fixed-income returns are least exposed to extreme loss scenarios.

8 Combined Portfolio Results

The combined 60/40 portfolio produces risk estimates sitting between equity and bond portfolios across every measure and both confidence levels, providing empirical validation of portfolio construction (Markowitz, 1952). At 99% confidence, historical ES of $23,826 represents a 36% reduction relative to the equity standalone of $37,161, greater than the 40% bond weighting alone would imply. This confirms a genuine diversification benefit beyond mechanical weighting, driven by imperfect positive correlation between the two assets.

The equity component dominates combined portfolio tail behaviour. Excess kurtosis of 5.666 means the normality assumption understates tail risk, evidenced by parametric VaR of $14,346 being the lowest estimate across all methods at 99%. The parametric

historical gap of $3,345 (23%) reflects the cost of normality. A risk manager relying solely on parametric estimates allocates $3,345 less capital per $1,000,000 invested than the distribution requires.

Complementary measures reveal meaningful but imperfect diversification. Annualised volatility falls 34.63% relative to equity but maximum drawdown of -17.13% is only 22.88% below equity's -22.21%. This asymmetry shows that volatility reduction overstates true tail risk reduction.

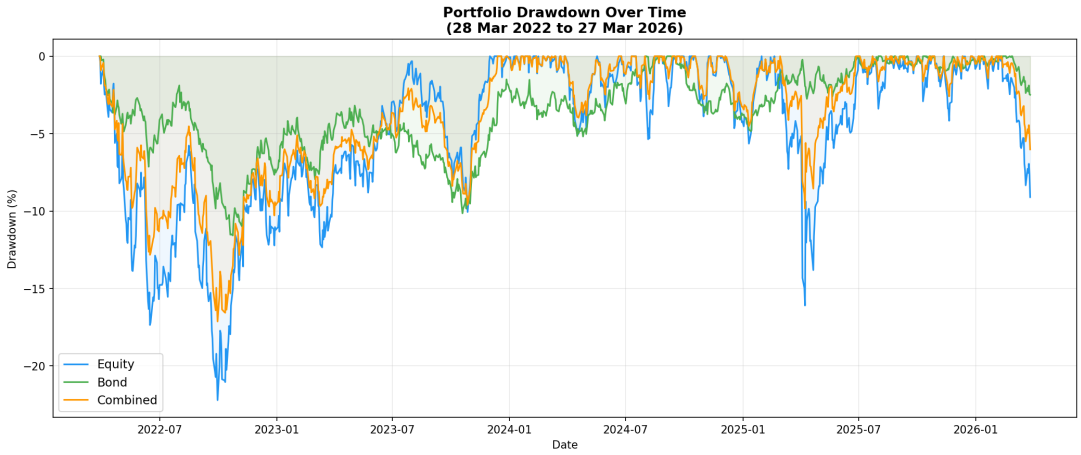


Figure 6: Portfolio Drawdown Over Time

Figure 6 illustrates why volatility reduction overstates true tail risk reduction. Both equity and bond components declined simultaneously during the 2022 stress period, directly failing the negative correlation assumption underlying the 60/40 construction. The combined portfolio's drawdown tracks equity more closely than bond throughout, confirming equity dominance. A secondary drawdown around April 2025 affects equity and combined but not bond, consistent with the equity-specific "Liberation Day" shock.

9 Comparison and Discussion

Across all three portfolios and both confidence levels, equity generates the highest risk estimates, bond the lowest and combined sits between them across every measure and method. As illustrated by the heatmap in Figure 7, no single methodology drives this result, strengthening our findings. The hierarchy is robust across all nine fundamentally different risk measures.

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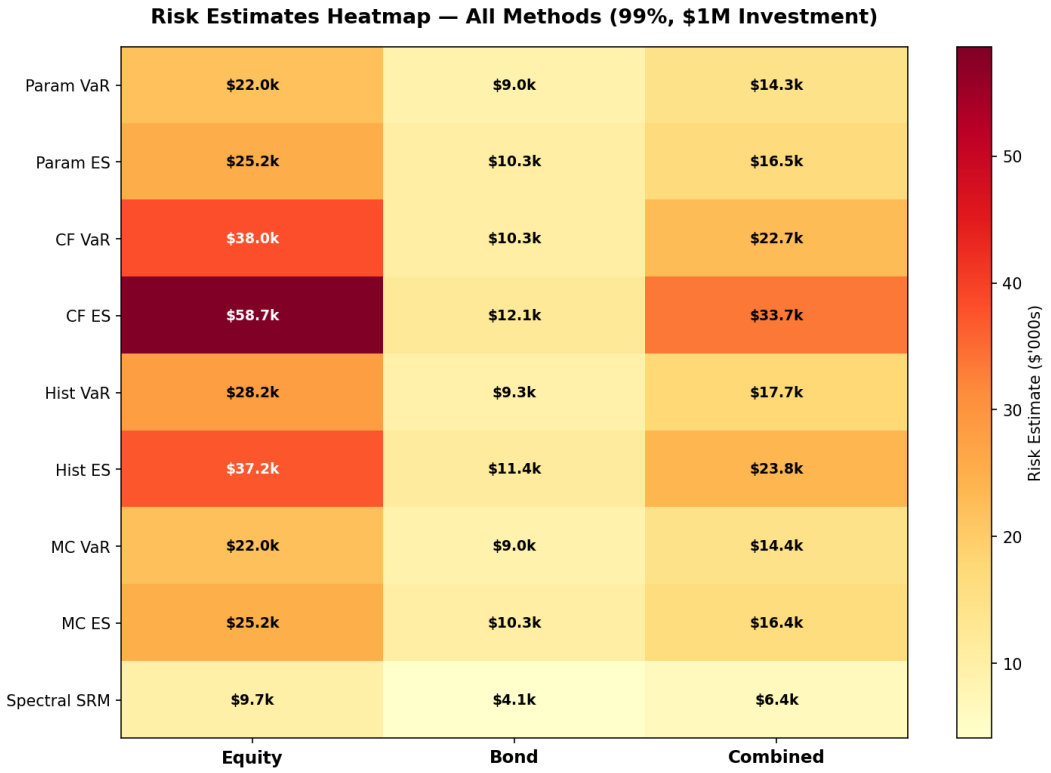


Figure 7: Risk Estimates Heatmap – All Methods at 99% Confidence

The most significant cross-method finding is how parametric-historical divergence scales with kurtosis. For equity, Historical VaR exceeds Parametric VaR by \$6,202, or 28.24%. For the combined portfolio, this narrows to \$3,345, or 23.3%, and collapses to \$269, or 2.98%, for bond. This gradient is in line with expectations of parametric methods being inversely related to excess kurtosis. A risk manager applying parametric methods uniformly would be undercapitalised for equity and combined risk but would obtain a reliable estimate for bond.

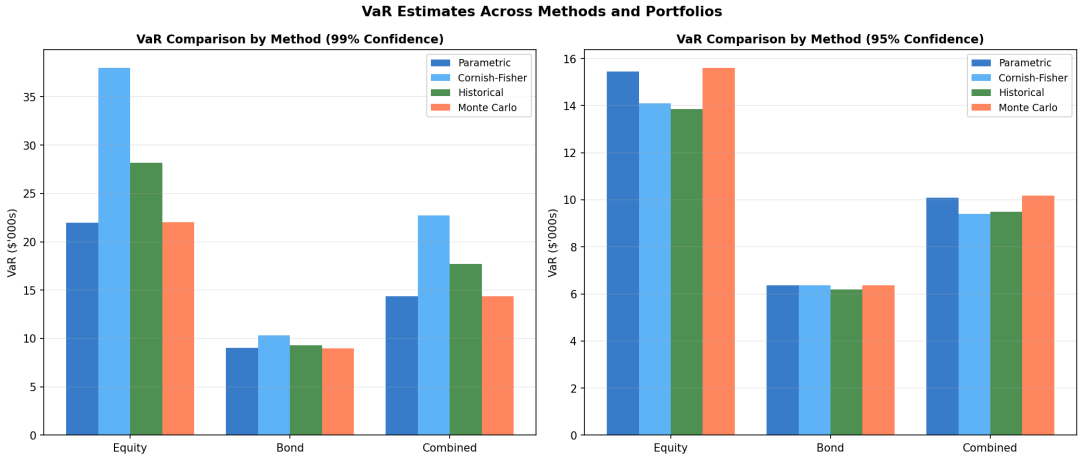


Figure 8: VaR Estimates Across Methods and Portfolios

Figure 8 illustrates this divergence, where CF partially addresses the underestimation of the parametric method by pushing VaR estimates materially higher at 99%. This is most pronounced for equity, where CF VaR dominates the Parametric VaR of $21,964, and least pronounced for bond, consistent with near-normality. At 95%, CF VaR falls below Parametric VaR for equity, indicating polynomial approximation breakdown under extreme kurtosis.

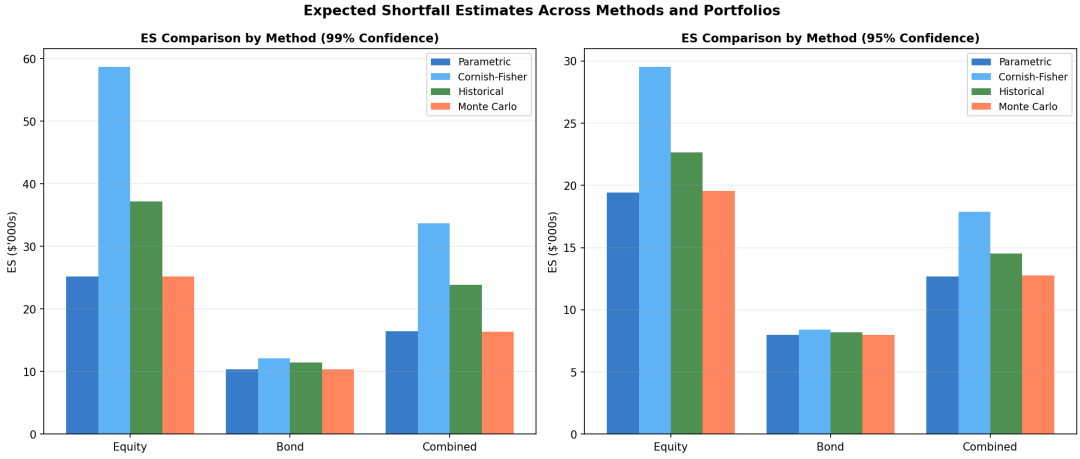


Figure 9: ES Estimates Across Methods and Portfolios

Figure 9 shows that bond ES estimates cluster tightly across all methods at both confidence levels, confirming that method choice is immaterial under near-normality. For equity and combined, CF ES at 95% is disregarded as it is inconsistent with every other method. This contrast illustrates why a one-size-fits-all approach to risk measurement is inappropriate across portfolios with different distributions. ES exceeds VaR without exception throughout, confirming internal consistency across all methods.

10 Backtesting Framework

10.1 Rolling Window Setup

A rolling backtesting framework was implemented to evaluate out-of-sample model reliability. Our initial estimation window spanning 28th March 2022 to 29th March 2024 is used to generate one-step ahead VaR and ES forecasts generated on a daily basis. The window is rolled forward recursively to produce a sequence of out-of-sample forecasts over the backtest period from 2nd April 2024 to 27th March 2026. This mirrors real-world risk management practice, where models adapt continuously to new information.

10.2 Evaluation Approach

VaR performance is evaluated using the Kupiec (1995) unconditional coverage test, assessing whether observed breach (portfolio loss exceeding predicted VaR) rates match expected rates. The Christoffersen (1998) independence test assesses whether breaches are clustered over time. Conditional coverage combines both dimensions. ES performance is evaluated by comparing forecasts against realised losses on VaR breach days. This assesses not only threshold accuracy but the severity of losses beyond it.

11 Backtesting Results

11.1 Value at Risk (VaR) Backtesting

Table 2: VaR Backtesting Summary at 99% Confidence

Portfolio	Method	Obs.	Breaches	Obs. Rate	Exp. Rate	Kupiec p	Indep. p	CC p	Kupiec	Indep.	CC
Equity	Parametric VaR	499	11	0.022	0.010	0.0196	0.2324	0.0322	Fail	Pass	Fail
Equity	Cornish-Fisher VaR	499	7	0.014	0.010	0.3939	0.0791	0.1489	Pass	Pass	Pass
Equity	Historical VaR	499	10	0.020	0.010	0.0473	0.1861	0.0584	Fail	Pass	Pass
Equity	Monte Carlo VaR	499	11	0.022	0.010	0.0196	0.2324	0.0322	Fail	Pass	Fail
Bond	Parametric VaR	499	3	0.006	0.010	0.3336	0.8488	0.6153	Pass	Pass	Pass
Bond	Cornish-Fisher VaR	499	2	0.004	0.010	0.1260	0.8989	0.3077	Pass	Pass	Pass
Bond	Historical VaR	499	3	0.006	0.010	0.3336	0.8488	0.6153	Pass	Pass	Pass
Bond	Monte Carlo VaR	499	3	0.006	0.010	0.3336	0.8488	0.6153	Pass	Pass	Pass
Combined	Parametric VaR	499	9	0.018	0.010	0.1049	0.1451	0.0929	Pass	Pass	Pass
Combined	Cornish-Fisher VaR	499	5	0.010	0.010	0.9964	0.0344	0.1067	Pass	Fail	Pass
Combined	Historical VaR	499	8	0.016	0.010	0.2131	0.1094	0.1279	Pass	Pass	Pass
Combined	Monte Carlo VaR	499	9	0.018	0.010	0.1049	0.1451	0.0929	Pass	Pass	Pass

For the equity portfolio, the parametric and Monte Carlo models generate 11 breaches (2.2%), more than double the expected 1% rate, leading to rejection under both the Kupiec and conditional coverage tests (Table 2). This confirms the previous finding that Gaussian-based models systematically underestimate tail risk. The CF approach performs best, with 7 breaches (1.4%), passing all tests, reflecting its ability to incorporate excess kurtosis. The Historical method lies between the two with 10 breaches, passing conditional coverage but slightly over-breaching.

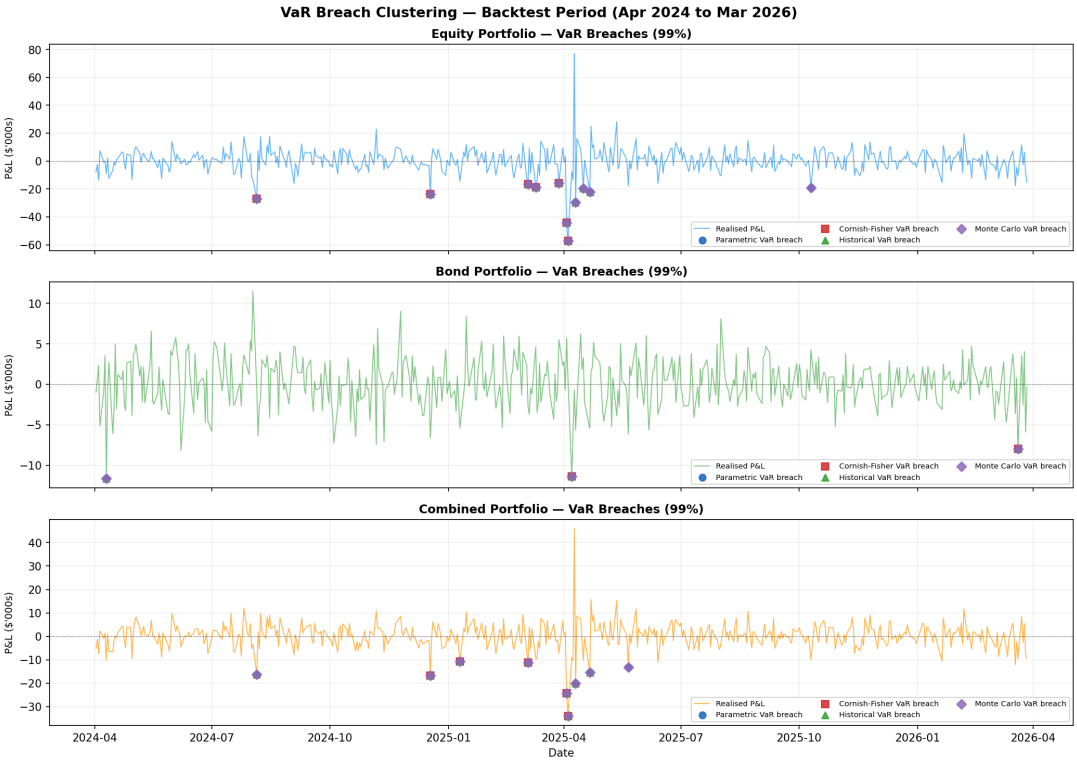


Figure 10: VaR Breach Clustering at 99% Confidence

Figure 10 confirms that equity breaches cluster around periods of market stress, particularly in early 2025. This highlights volatility clustering not captured by static models. For the bond portfolio, all methods produce only 2–3 breaches with all tests passed, consistent with near-normal return distributions and an absence of clustering. For the combined portfolio, breaches range from 5 to 9 across methods, with CF showing mild dependence issues (p-value = 0.0344), driven by equity-aligned stress periods.



Figure 11: Rolling One-Day-Ahead VaR Forecasts at 99% Confidence

Figure 11 shows Parametric and Monte Carlo estimates moving almost identically across all portfolios, confirming their shared Gaussian assumption. CF consistently produces higher VaR estimates, reflecting higher-moment sensitivity, with a pronounced spike around early 2025 visible across all methods. This demonstrates that the rolling framework adapts to changing market conditions, although differences across models remain persistent.

11.2 Expected Shortfall (ES) Performance

Table 3: ES Backtesting Results at 99% Confidence

Portfolio	Method	ES MSE	Bias	Exceedances
Bond	Parametric VaR	1.020858×10^6	-354.733469	3
Bond	Monte Carlo VaR	1.028003×10^6	-342.085976	3
Bond	Historical VaR	1.694794×10^6	159.298638	3
Bond	Cornish-Fisher VaR	2.095790×10^6	222.608081	2
Combined	Historical VaR	6.649908×10^7	-2176.224686	8
Combined	Monte Carlo VaR	7.764243×10^7	-4847.230499	9
Combined	Parametric VaR	7.827476×10^7	-4919.032673	9
Combined	Cornish-Fisher VaR	8.406457×10^7	-5030.246933	5
Equity	Cornish-Fisher VaR	1.798193×10^8	-4455.222551	7
Equity	Historical VaR	2.122634×10^8	-1032.038573	10
Equity	Monte Carlo VaR	2.158288×10^8	-7014.866108	11
Equity	Parametric VaR	2.170666×10^8	-7018.346612	11

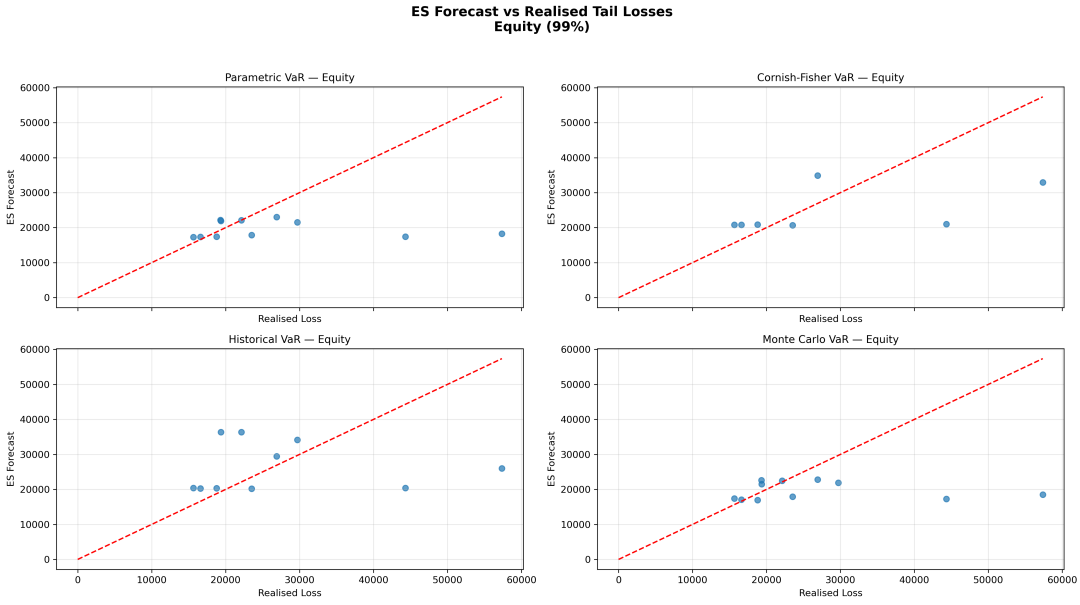


Figure 12: ES Forecast vs Realised Losses for Equity at 99% Confidence

For equity, all models exhibit negative bias, indicating systematic underestimation of tail losses, as shown in Table 3. Parametric and Monte Carlo perform worst with the largest bias, approximately -$7,000, and the highest MSE. The CF model achieves the lowest MSE and ranks best overall, while the Historical method provides the smallest bias, indicating more accurate estimation of realised losses. Figure 12 confirms forecasts lie below the 45-degree line throughout.

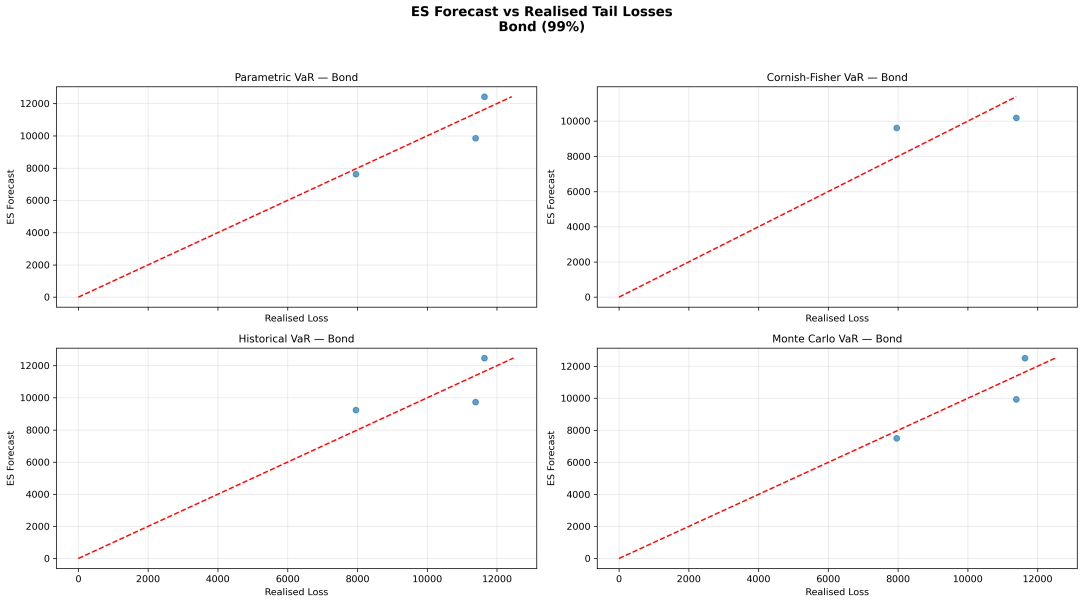


Figure 13: ES Forecast vs Realised Losses for Bond at 99% Confidence

For bond, ES performance is strong across all methods, with bias close to zero and significantly lower MSE than equity, as shown in Table 3. Parametric performs best, consistent with near-normality. Figure 13 shows that points lie close to the 45-degree line, indicating accurate alignment between predicted and realised losses.

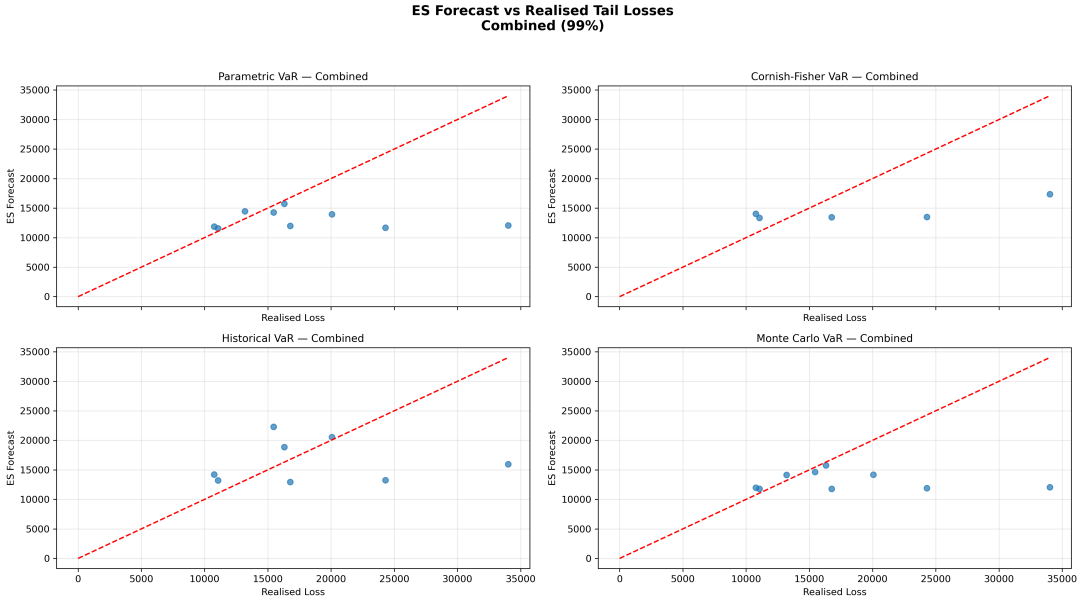


Figure 14: ES Forecast vs Realised Losses for Combined Portfolio at 99% Confidence

For the combined portfolio, Historical achieves the lowest MSE and relatively small bias compared to other models, as shown in Table 3. Parametric and Monte Carlo methods again underestimate losses more significantly, while Cornish-Fisher slightly over-adjusts. Figure 14 confirms intermediate deviation between the equity and bond patterns.

12 Sensitivity Analysis

12.1 Confidence Level Sensitivity: 95% vs 99%

Table 4: VaR Backtesting Summary at 95% Confidence

Portfolio	Method	Obs.	Breaches	Obs. Rate	Exp. Rate	Kupiec p	Indep. p	CC p	Kupiec	Indep.	CC
Equity	Parametric VaR	499	22	0.0441	0.0500	0.5367	0.0676	0.1556	Pass	Pass	Pass
Equity	Cornish-Fisher VaR	499	29	0.0581	0.0500	0.4169	0.0204	0.0489	Pass	Fail	Fail
Equity	Historical VaR	499	26	0.0521	0.0500	0.8303	0.0402	0.1191	Pass	Fail	Pass
Equity	Monte Carlo VaR	499	22	0.0441	0.0500	0.5367	0.0676	0.1556	Pass	Pass	Pass
Bond	Parametric VaR	499	12	0.0240	0.0500	0.0032	0.4414	0.0097	Fail	Pass	Fail
Bond	Cornish-Fisher VaR	499	12	0.0240	0.0500	0.0032	0.4414	0.0097	Fail	Pass	Fail
Bond	Historical VaR	499	9	0.0180	0.0500	0.0002	0.5649	0.0007	Fail	Pass	Fail
Bond	Monte Carlo VaR	499	11	0.0220	0.0500	0.0013	0.4808	0.0045	Fail	Pass	Fail
Combined	Parametric VaR	499	24	0.0481	0.0500	0.8444	0.0215	0.0699	Pass	Fail	Pass
Combined	Cornish-Fisher VaR	499	26	0.0521	0.0500	0.8303	0.0070	0.0259	Pass	Fail	Fail
Combined	Historical VaR	499	25	0.0501	0.0500	0.9918	0.0297	0.0941	Pass	Fail	Pass
Combined	Monte Carlo VaR	499	24	0.0481	0.0500	0.8444	0.0215	0.0699	Pass	Fail	Pass

Table 5: ES Performance Ranking at 95% Confidence

Portfolio	Method	ES MSE	Bias	Exceedances
Bond	Monte Carlo VaR	2.968778×10^6	84.467278	11
Bond	Parametric VaR	3.030258×10^6	207.249964	12
Bond	Historical VaR	3.191650×10^6	-205.906892	9
Bond	Cornish-Fisher VaR	3.395661×10^6	407.126763	12
Combined	Historical VaR	4.226448×10^7	-1252.719152	25
Combined	Monte Carlo VaR	4.447026×10^7	-2483.091786	24
Combined	Parametric VaR	4.448351×10^7	-2513.976189	24
Combined	Cornish-Fisher VaR	8.181739×10^7	3479.984441	26
Equity	Historical VaR	1.244004×10^8	-2383.652764	26
Equity	Parametric VaR	1.496607×10^8	-5443.773741	22
Equity	Monte Carlo VaR	1.500140×10^8	-5397.633158	22
Equity	Cornish-Fisher VaR	2.972453×10^8	7792.120341	29

Repeating the backtesting framework at 95% confidence reveals how model rankings shift with confidence level selection.

For equity, all models produce breach rates close to the expected 5% at 95%, passing the Kupiec test, contrasting the 99% results where parametric and MC doubled the expected breach rate. However, CF and Historical both fail the independence test at 95%, indicating breach clustering. This reversal is reinforced by ES results. CF performs worst at 95% with the highest MSE and positive bias, while Historical provides the lowest MSE. The deterioration of CF at 95% is consistent with the previous finding that the

polynomial approximation becomes unreliable under extreme kurtosis at lower confidence levels.

For bond, all models significantly under-breach at 95% (only 9-12 breaches), leading to Kupiec rejection (Table 4). VaR estimates are overly conservative at 95% though models err on the side of caution rather than calibration error. ES performance remains relatively stable (Table 5), with small biases and similar MSE values across methods. For near-normal distributions, model differences are less pronounced but calibration becomes sensitive to the chosen confidence level.

For combined, results are broadly stable across confidence levels with breach rates close to expectations and pass coverage tests (Table 4), although independence failures are more frequent, suggesting mild clustering effects. ES results again favour the Historical approach, delivering lowest MSE and moderate bias, while parametric and MC continue to underestimate losses.

12.2 Sensitivity to Model Assumptions

Parametric and MC methods produce nearly identical results across all portfolios and confidence levels, confirming their shared Gaussian assumption leads to consistent tail risk underestimation for equity. CF shows greater variability, effective at 99% but deteriorating at 95%, indicating instability in moment-based adjustments. Historical remains the most stable across all settings as it imposes no distributional assumptions.

12.3 Time-Varying Risk and Model Responsiveness

Figure 11 illustrates that all models adjust to changing volatility over time, with a pronounced VaR spike in early 2025. However, CF responds most aggressively while Historical adjust gradually, confirming that model choice affects not only the level of risk estimates but the speed at which risk is updated during stress. No single model is universally optimal and method selection must reflect both portfolio distributional characteristics and the chosen confidence level.

13 Comparison, Model Recommendations and Additional Considerations

This report's recommendations draw on both full-sample distributional analysis and rolling backtesting results. Full-sample analysis identifies which models are most ap-

propriate given each portfolio's return distribution, while backtesting evaluates whether those models produce reliable out-of-sample risk forecasts.

Across both analyses, the risk hierarchy is consistent. Equity is most risky, bond least, and the combined lies between them. This confirms that portfolio composition is the primary driver of risk, while model choice determines measurement accuracy.

For the equity portfolio, Historical ES is recommended as the primary tail-risk measure, supported by CF VaR for threshold monitoring. Excess kurtosis of 7.149 means parametric methods systematically understate extreme losses, making Historical ES the theoretically appropriate measure. Backtesting confirms CF VaR performs best at the 99% level, passing the VaR backtesting criteria. Therefore, Historical ES should guide capital allocation, while Cornish-Fisher VaR is monitored for breach frequency.

For the bond portfolio, Parametric ES is recommended with Parametric VaR retained for validation. Kurtosis of 1.141 confirms near-normality, the parametric-historical gap collapses to $269 and backtesting shows strong model convergence across all methods. The parametric approach is preferred for its accuracy, simplicity and interpretability for the Board. It is noted that backtesting reveals VaR models are overly conservative at 95%, though this represents over-estimation rather than calibration error and does not alter the recommendation.

For the combined portfolio, Historical ES is recommended as the primary measure, supported by Historical VaR for backtesting. Despite diversification benefits, tail behaviour remains equity-driven. Historical methods are therefore preferable because they capture realised empirical tail losses without imposing restrictive distributional assumptions, a finding supported by both distributional analysis and backtesting performance.

The Board should also note broader implications from this analysis. The $6,202 capital shortfall per $1,000,000 of equity exposure between parametric and Historical VaR is a consequence of fat tails (not modelling) and should inform capital allocation decisions. The 2022 stress period demonstrated that the negative equity-bond correlation underlying the 60/40 construction can fail when diversification is most needed. The April 2025 Liberation Day shock illustrates that geopolitical tail events can generate losses no model fully anticipates. All estimates assume a one-day horizon and static portfolio weights. Thus, multi-day risk figures derived from square root of time scaling should be treated with caution given the normality assumption this report has empirically rejected.

ES is the recommended primary metric across all portfolios. It captures loss severity beyond the threshold which VaR cannot. Crucially, model selection should be tailored to each portfolio's distributional characteristics rather than applied uniformly.

14 Conclusion

This report presents a risk measurement analysis across three $1,000,000 portfolios: equity, bond and combined. These were assessed using four VaR and ES methodologies, supplemented by an Exponential Spectral Risk Measure, evaluated using a rolling backtesting framework.

The central finding is that no single model is universally optimal. Performance varies by portfolio characteristics and confidence level. For equity and combined, excess kurtosis of 7.149 and 5.666 respectively causes Gaussian-based methods to systematically understate tail risk. This was confirmed by both distributional analysis and backtesting. For bond, near-normality means parametric methods are reliably calibrated and model choice is far less consequential.

This report recommends Historical ES as the primary risk measure for equity and combined portfolios, supported by CF VaR for breach monitoring and Parametric ES for the bond portfolio. Across all portfolios, ES is preferred over VaR as the capital allocation metric. It captures the severity of losses beyond the VaR threshold which is the most critical information for this Board Risk Committee.

Generative AI Statement

This report uses Claude AI to help develop Python code especially related to auxiliary graphs. However, this was thoroughly examined and edited by the group and all report writing is our own work.

Appendix

Appendix 1: DJIA Constituents

Table 6: DJIA Constituents

Ticker	**Name**
GS UN Equity	Goldman Sachs Group
CAT	Caterpillar Inc
MSFT	Microsoft Corp
AMGN	Amgen Inc
HD	Home Depot Inc
SHW	Sherwin-Williams Co
AXP	American Express Co
MCD	McDonald's Corp
JPM	JP Morgan Chase & Co
UNH	UnitedHealth Group Inc
V	Visa Inc
TRV	Travelers Cos Inc
AAPL	Apple Inc
JNJ	Johnson & Johnson
IBM	International Business Machine
HON	Honeywell International Inc
AMZN	Amazon.com Inc
BA	Boeing Co
CVX	Chevron Corp
NVDA	NVIDIA Corp
CRM	Salesforce Inc
MMM	3M
PG	Procter & Gamble Co
WMT	Walmart Inc
MRK	Merck & Co Inc
DIS	Walt Disney Co
CSCO	Cisco Systems Inc
KO	Coca-Cola Co
VZ	Verizon Communications Inc
NKE	NIKE Inc

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